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INSO4998 – Undergraduate Research in Software Engineering
Quantum Cryptography

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Goal:

With the PR-LASMP event approaching, and after receiving the green light for the abstract, I set out to begin the construction of the presentation that is going to be presented at the event.

Actions:

Juan D. Guadalupe Rosado

Present:

- Began preparation of the presentation that is going to be used for the PR-LASMP
    - Might also draft a poster if time permits (and if we register for it)

- Began reading "Is Circuit Depth Accurate for Comparing Quantum Circuit Runtimes?" by Matthew Tremba, Paul Hovland, and Ji Liu, in a bid to understand how different metrics are used and considered in Quantum circuits.

Future:

- Begin the construction of the arithmetic circuit discussed in previous meetings.
- Construct a classical version of the above circuit that is going to be used as comparison.
- After PR_LSAMP materials are done and ready, delve deeper into the optimization of quantum circuits.

Daniel C. Colón Rivera

Present:

- Created an account on IBM Quantum Platform, created an API key and successfully ran a basic circuit on quantum hardware.

Future:

- Run our current algorithm circuits on IBM Quantum.
- Read "Is Circuit Depth Accurate for Comparing Quantum Circuit Runtimes?".
- Devise an appropriate metric that is both applicable to quantum hardware and permits comparison between classical and quantum performance.
- Come up with a problem that makes extensive use of arithmetic circuits and has the potential to demonstrate quantum advantage.